

MAH, Commercialization Frictions, and Original-Drug Innovation

Abstract

This paper develops a partial-equilibrium mechanism model of how China's Marketing Authorization Holder (MAH) system can affect original-drug innovation. Firms choose original-drug R&D before successful opportunities are realized. After a successful opportunity arrives, a firm chooses whether to commercialize internally, retain authorization while entrusting production to a qualified manufacturer, transfer or out-license the opportunity, or abandon it. MAH affects expected R&D value by adding a regulated holder-producer separation route, lowering the route-specific policy and contractual friction of that retained external route, and changing the composite probability that a latent opportunity becomes an observed approval or launch. The baseline closes only the qualified contract-manufacturing service market, generating an endogenous service price $p_m^*(M)$ that can attenuate the reform's effect. The model does not treat MAH as a demand shock, a scientific-productivity shock, a removal of holder responsibility, or a universal reduction in all commercialization costs. It delivers comparative statics for route value, R&D intensity, and observed realized original-drug outcomes, and it clarifies which data moments can discipline composite objects in a mechanism calibration.

Keywords: MAH; original-drug innovation; commercialization frictions; entrusted production; route choice.

JEL codes: D22; L65; O31; O38; I18.

1 Introduction

China's MAH system separates the right to hold marketing authorization from a strict requirement of in-house production. This institutional change is potentially important for original-drug innovation because many research-originating firms can generate valuable projects but lack complete manufacturing capacity. The central mechanism studied here is therefore not demand expansion or scientific productivity. It is an expected-commercialization-value channel: when a firm anticipates that a future successful project

can be commercialized through an additional retained route, current R&D becomes more valuable.

The mechanism builds on two established ideas. First, pharmaceutical R&D responds to expected downstream rewards (Schmookler, 1966; Acemoglu and Linn, 2004; Finkelstein, 2004; Dubois et al., 2015; Budish et al., 2015). Second, an innovator’s ability to appropriate returns depends on complementary assets and commercialization strategy (Teece, 1986; Arora et al., 2001; Gans and Stern, 2003). MAH fits this second logic because it changes how an innovator may access manufacturing capability while retaining the authorization asset.

The model is deliberately narrow. It is not a full dynamic structural model of the pharmaceutical industry. It does not solve product-market clearing, R&D input-market clearing, free entry, exit, or an invariant firm distribution. The baseline closes only the market directly tied to the MAH mechanism: the market for qualified contract-manufacturing services. This single closure is important because MAH-induced demand for entrusted production can raise the service price $p_m^*(M)$, preventing the model from mechanically implying that MAH increases innovation for all firms.

The contribution is to organize the mechanism in a way that is internally consistent with feasible data. The theory separates latent opportunity arrival from observed approvals or launches, keeps residual holder responsibility in the entrusted-route payoff, and treats route-use data as a proxy for realized holder-producer separation rather than as the policy treatment itself. The empirical role of the model is correspondingly modest: it maps feasible approval-side moments to composite objects such as route-realization value and innovation-realization scale, without claiming to identify every primitive friction separately.

The model is presented in two layers. The main text gives the shortest complete version needed for the paper’s argument. The technical appendix gives the object-status audit, payoff accounting, Bellman derivation, CMO market regularity conditions, comparative-static derivations, and calibration boundaries. Research notes are kept outside the manuscript and are used only as a model-design control panel.

2 Institutional Background

The relevant institutional point is timing. Under China’s drug-registration framework, a firm need not already be the final commercial holder when it starts R&D. The economic choice is forward-looking: a firm invests today while anticipating the commercialization environment it will face if a project succeeds. After obtaining a Drug Approval License, the applicant becomes the MAH (National Medical Products Administration, 2022a). The model therefore treats MAH as affecting expected future commercialization feasibility

rather than current scientific productivity.

The MAH reform creates a regulated holder-producer separation route. The 2016 pilot allowed eligible applicants in selected regions to become holders under the pilot arrangement ([General Office of the State Council of the People's Republic of China, 2016](#); [China Food and Drug Administration, 2016](#)). The later legal framework consolidated the principle that an MAH may manufacture by itself or entrust production to a qualified manufacturer ([National Medical Products Administration, 2019](#)). Subsequent NMPA rules require holder-side quality responsibility, assessment of the entrusted manufacturer, quality and entrustment agreements, and ongoing supervision ([National Medical Products Administration, 2022c,b, 2023](#)).

These rules imply two modeling restrictions. First, entrusted production is not anonymous outsourcing. It is a regulated retained route in which authorization remains with the holder while production is performed by a qualified external manufacturer. Second, MAH does not eliminate holder responsibility. The holder retains quality, safety, lifecycle, and monitoring obligations. The model captures this through a residual entrusted-route burden μ_i^E and a composite realization probability $\zeta_i^E(M, p_m)$.

3 Model

3.1 Environment and Timing

There are two periods within the firm problem. In period t , firm i chooses original-drug R&D effort x_i . In period $t + 1$, successful original-drug opportunities arrive, and the firm chooses a commercialization route for each opportunity. The institutional regime is $M \in \{0, 1\}$, where $M = 1$ denotes the MAH regime.

Firm characteristics are summarized by

$$\theta_i = (a_i, k_i, h_i^I, q_i^E, \mu_i^E, S_i, \tau_i^T),$$

where a_i is research productivity, k_i indexes internal manufacturing capability, h_i^I indicates whether internal production is feasible, q_i^E indicates whether the opportunity is feasible for entrusted production, μ_i^E is the residual holder-side burden under entrusted production, S_i is the gross non-retained transfer or out-licensing value, and τ_i^T is the transfer-route friction.

The model is partial equilibrium except for the qualified contract-manufacturing service market. It takes the market-return environment and firm-characteristic distributions as given. The service price $p_m^*(M)$ is determined by market clearing:

$$D_m(p_m; M) = S_m(p_m). \tag{1}$$

3.2 Route Set and Route Payoffs

Conditional on a successful opportunity, the route set is

$$\mathcal{R}_i(M) = \{A, T\} \cup \{I : h_i^I = 1\} \cup \{E : M = 1, q_i^E = 1\}. \quad (2)$$

Here I is internal production, E is retained entrusted production, T is a non-retained transfer, sale, or out-licensing option, and A is abandonment or delay.

Let R_i^{event} denote the route-independent commercialization-event return from a realized original drug, and let v be the continuation value of adding one retained product or certificate to the firm's stock. The entrusted-route payoff is

$$G_i^E(M, p_m) = \zeta_i^E(M, p_m) [\bar{R}_i^E + v] - p_m - \tau_i^E(M) - \mu_i^E, \quad (3)$$

where $\tau_i^E(M)$ is the MAH-reducible policy and contractual friction, and

$$\zeta_i^E(M, p_m) = \zeta^E(\tau_i^E(M), q_i^E, \theta_i, p_m)$$

is a composite probability linking latent opportunity arrival to observed approval, launch, or realized retained outcome. The baseline uses $\bar{R}_i^E = \mathbb{E}[R_i \mid L_i = 1, E]$ as a compressed post-production payoff object.

The other route payoffs are

$$G_i^I = R_i^{event} + v - C^I(k_i), \quad (4)$$

$$G_i^T = S_i - \tau_i^T, \quad (5)$$

$$G_i^A = 0. \quad (6)$$

The transfer value $S_i - \tau_i^T$ is kept exogenous in the baseline. A Grossman-Hart-Moore-style outside-option interpretation is useful, but it is not added as a main state variable ([Grossman and Hart, 1986](#); [Hart and Moore, 1990](#)).

3.3 Route Choice and CMO Market Closure

Route choice follows a logit model throughout:

$$P_i(r \mid M, p_m) = \frac{\exp(G_i^r(M, p_m)/\sigma_r)}{\sum_{\ell \in \mathcal{R}_i(M)} \exp(G_i^\ell(M, p_m)/\sigma_r)}, \quad \sigma_r > 0. \quad (7)$$

The inclusive value of a successful opportunity is

$$\Gamma_i(M, p_m) = \sigma_r \log \sum_{r \in \mathcal{R}_i(M)} \exp(G_i^r(M, p_m)/\sigma_r). \quad (8)$$

The deterministic maximum is the limiting case as $\sigma_r \rightarrow 0$, not the baseline model.

Demand for qualified entrusted manufacturing services is generated only by realized retained entrusted production:

$$D_m(p_m; M) = \int a_i x_i^*(M, p_m) P_i(E \mid M, p_m) \zeta_i^E(M, p_m) dH_i. \quad (9)$$

Qualified producer supply is upward sloping,

$$S_m(p_m) = \int s(p_m, z_j^C) dH_C(z_j^C), \quad \frac{\partial s(p_m, z_j^C)}{\partial p_m} \geq 0. \quad (10)$$

The equilibrium service price $p_m^*(M)$ solves equations (1)–(10). No other market is closed.

3.4 R&D Choice and Observed Outcomes

R&D effort produces latent successful original-drug opportunities according to

$$\lambda_i^{lat} = a_i x_i, \quad (11)$$

with cost

$$C_i^R(x_i) = \frac{\kappa}{2} x_i^2, \quad \kappa > 0. \quad (12)$$

Because each latent success has expected route value $\Gamma_i(M, p_m^*(M))$, the firm solves

$$\max_{x_i \geq 0} \beta a_i x_i \Gamma_i(M, p_m^*(M)) - \frac{\kappa}{2} x_i^2.$$

The interior solution is

$$x_i^*(M) = \frac{\beta a_i}{\kappa} \Gamma_i(M, p_m^*(M)). \quad (13)$$

Observed realized output differs from latent opportunity arrival:

$$\lambda_i^{obs}(M) = a_i x_i^*(M) \sum_{r \in \mathcal{R}_i(M)} P_i(r \mid M, p_m^*(M)) \zeta_i^r(M, p_m^*(M)). \quad (14)$$

For the retained entrusted route,

$$\lambda_i^{obs,E}(M) = a_i x_i^*(M) P_i(E \mid M, p_m^*(M)) \zeta_i^E(M, p_m^*(M)). \quad (15)$$

4 Theoretical Implications

Proposition 1 (Logit-weighted R&D response). *For a firm with $E \in \mathcal{R}_i(M)$, holding p_m , ζ_i^E , and other route payoffs fixed,*

$$\frac{\partial x_i^*}{\partial \tau_i^E} = -\frac{\beta a_i}{\kappa} P_i(E \mid M, p_m) < 0. \quad (16)$$

The effect of a lower entrusted-route friction is larger when the firm assigns higher route probability to retained entrusted production.

The result follows because the derivative of the log-sum inclusive value with respect to a route payoff equals that route’s logit probability. The model therefore predicts probability-weighted sorting rather than a universal response. Firms for which the entrusted route is infeasible, unattractive, or dominated by internal production respond little.

The technical appendix also gives the full derivative when ζ_i^E varies with the entrusted-route friction. The main-text expression isolates the direct route-friction channel by holding realization probability fixed.

Proposition 2 (Observed entrusted-route decomposition). *The change in observed retained entrusted output can be decomposed as*

$$\begin{aligned} d\lambda_i^{obs,E} &= a_i P_i(E \mid M, p_m^*(M)) \zeta_i^E(M, p_m^*(M)) dx_i^* \\ &\quad + a_i x_i^*(M) \zeta_i^E(M, p_m^*(M)) dP_i(E \mid M, p_m^*(M)) \\ &\quad + a_i x_i^*(M) P_i(E \mid M, p_m^*(M)) d\zeta_i^E(M, p_m^*(M)). \end{aligned} \quad (17)$$

Observed approvals or launches may rise because firms invest more in R&D, because successful projects are more likely to use the entrusted route, or because the composite realization probability improves. Approval-side data do not separately identify these components without additional application, status, and project-level information.

Proposition 3 (CMO price attenuation). *Suppose MAH lowers τ_i^E for a positive-measure set of firms and raises demand for entrusted production. If qualified CMO supply is upward sloping, the induced increase in $p_m^*(M)$ partially offsets the entrusted-route value gain.*

The entrusted-route payoff depends on the net wedge $p_m^*(M) + \tau_i^E(M) + \mu_i^E$. A reduction in policy or contracting friction increases route value only to the extent that it is not fully absorbed by a higher equilibrium service price or by residual holder-side responsibilities. This is the main force preventing a mechanical “MAH raises innovation for everyone” conclusion.

Corollary 1 (Aggregate implication). *For a fixed firm distribution, aggregate latent original-drug opportunities rise when the positive $\Gamma_i(M, p_m^*(M))$ response among treated-relevant firms is not offset by CMO price feedback and when the affected set has positive measure. Aggregate observed original-drug outcomes additionally require that route-use and realization-probability terms do not fall enough to offset the latent-opportunity gain. The result is a private-value and realized-output implication, not a welfare theorem.*

5 Empirical and Calibration Implications

The model is intended to discipline a mechanism calibration, not to identify every primitive friction. The key empirical distinction is between latent opportunity arrival and observed realized outcomes. Approval data, holder-producer links, and original-drug counts are close to realized outcomes. They are not direct observations of latent ideas, project-level scientific success, or true route-specific legal and managerial costs.

Table 1: Model Objects and Feasible Data Moments

Model object	Feasible moment	Claim boundary
$P_i(E \mid M, p_m^*(M)) \zeta_i^E(M, p_m^*(M))$	Holder-producer split among realized original drugs	Disciplines a composite route-realization wedge, not a primitive τ_i^E or μ_i^E separately.
$a_i x_i^*(M)$ and realization scale	Realized original-drug counts	Disciplines innovation-realization scale, not latent idea arrival alone.
$p_m^*(M)$ or CMO scarcity	Qualified entrusted-producer participation or capacity proxy	Needed to interpret attenuation; direct CMO prices may be unavailable.
Entry-cost distribution F_e	Firm-year research-oriented entry panel	Future or sensitivity module unless credible entry data are merged.

Three calibration boundaries follow. First, route-use moments should be interpreted as evidence about relative route attractiveness among realized products, not as a direct estimate of the MAH policy treatment. Second, original-drug counts discipline a composite scale combining R&D effort, research productivity, route choice, and realization probabilities. Third, entry should not be a baseline calibrated outcome unless the data contain a credible firm-year entry panel and pre-reform capability measures.

The baseline empirical exercise should therefore focus on approval-side realized-product moments: realized original-drug counts, original-drug shares, holder-producer splits, and heterogeneity by pre-reform manufacturing capability where available. Application-to-approval conversion, true project-level success probabilities, CMO prices, and research-oriented entry are natural future modules, but they should not be treated as already identified by approval-side records alone.

The formal appendix documents the corresponding moment equations, route-share log-odds mapping, and entry-module boundary. This separation is intentional: the main paper states what the feasible moments can support, while the appendix records the algebra behind that claim.

6 Conclusion

The paper’s main mechanism is simple: MAH changes the value of R&D by changing the commercialization route set available after a successful original-drug opportunity arrives. The retained entrusted route matters most for firms that can generate projects but lack complete internal manufacturing capability, and its value depends on route feasibility, residual holder responsibility, composite realization probability, outside options, and the equilibrium price of qualified contract-manufacturing services. The model remains intentionally partial equilibrium. This restriction is useful because it keeps the theory aligned with the paper’s empirical objective: explaining how a specific institutional change can alter expected route value, R&D incentives, and observed realized original-drug outcomes without turning the project into a full industry equilibrium or welfare model.

References

- Acemoglu, Daron and Joshua Linn**, “Market Size in Innovation: Theory and Evidence from the Pharmaceutical Industry,” *Quarterly Journal of Economics*, 2004, 119 (3), 1049–1090.
- Arora, Ashish, Andrea Fosfuri, and Alfonso Gambardella**, *Markets for Technology: The Economics of Innovation and Corporate Strategy*, Cambridge, MA: MIT Press, 2001.
- Budish, Eric, Benjamin N. Roin, and Heidi Williams**, “Do Firms Underinvest in Long-Term Research? Evidence from Cancer Clinical Trials,” *American Economic Review*, 2015, 105 (7), 2044–2085.
- China Food and Drug Administration**, “Notice on Matters Related to the Marketing Authorization Holder System Pilot,” <https://www.ccfdie.org/zryyxxw/zcfg/fgdt/webinfo/2016/07/1466184737348633.htm> 2016. Accessed 2026-05-20.
- Dubois, Pierre, Olivier de Mouzon, Fiona Scott Morton, and Paul Seabright**, “Market Size and Pharmaceutical Innovation,” *RAND Journal of Economics*, 2015, 46 (4), 844–871.
- Finkelstein, Amy**, “Static and Dynamic Effects of Health Policy: Evidence from the Vaccine Industry,” *Quarterly Journal of Economics*, 2004, 119 (2), 527–564.
- Gans, Joshua S. and Scott Stern**, “The Product Market and the Market for Ideas: Commercialization Strategies for Technology Entrepreneurs,” *Research Policy*, 2003, 32 (2), 333–350.
- General Office of the State Council of the People’s Republic of China**, “Pilot Plan for the Marketing Authorization Holder System,” https://www.gov.cn/zhengce/content/2016-06/06/content_5079954.htm 2016. Accessed 2026-05-20.
- Grossman, Sanford J. and Oliver D. Hart**, “The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration,” *Journal of Political Economy*, 1986, 94 (4), 691–719.
- Hart, Oliver and John Moore**, “Property Rights and the Nature of the Firm,” *Journal of Political Economy*, 1990, 98 (6), 1119–1158.
- National Medical Products Administration**, “Drug Administration Law of the People’s Republic of China,” https://english.nmpa.gov.cn/2019-09/26/c_773012.htm 2019. Accessed 2026-05-18.

- , “Provisions for Drug Registration,” https://english.nmpa.gov.cn/2022-06/30/c_785628.htm 2022. Accessed 2026-05-18.
 - , “Provisions for the Supervision and Administration of Drug Manufacturing,” https://english.nmpa.gov.cn/2022-06/30/c_785630_2.htm 2022. Accessed 2026-05-18.
 - , “Provisions on Supervising the Implementation of the Drug Marketing Authorization Holder’s Primary Responsibility for Drug Quality and Safety,” <https://www.nmpa.gov.cn/xxgk/fgwj/xzhgfxwj/20221229192548121.html> 2022. Released 2022-12-29; effective 2023-03-01; accessed 2026-05-20.
 - , “Announcement on Strengthening Supervision and Administration of Entrusted Production by Drug Marketing Authorization Holders,” <https://www.nmpa.gov.cn/xxgk/fgwj/xzhgfxwj/20231023160426145.html> 2023. NMPA Announcement No. 132 of 2023; accessed 2026-05-20.
- Schmookler, Jacob**, *Invention and Economic Growth*, Cambridge, MA: Harvard University Press, 1966.
- Teece, David J.**, “Profiting from Technological Innovation: Implications for Integration, Collaboration, Licensing and Public Policy,” *Research Policy*, 1986, 15 (6), 285–305.